

A Multi-Stream Temporal Context Model for Narrative Memory

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Abstract

The Temporal Context Model⁹ and its extension the Context Maintenance and Retrieval model¹⁵ together provide a widely-used computational framework for understanding how memory for a sequence of events depends on the temporal context in which they were encoded. Cornell and Zhang⁵ recently extended CMR with a slower list-level *hierarchical context* that synchronizes to the item-level context at list boundaries and captures between-list structure in free recall. Here we propose a *Multi-Stream Temporal Context Model* (MS-TCM) that extends Cornell and Zhang’s (2025) hierarchical CMR with exactly one additional mechanism: *storyline-return reinstatement at encoding* with strength λ . When a listener or viewer interleaves multiple storylines and later returns to a previously-abandoned storyline, the corresponding storyline-level context is partially reinstated from a cached snapshot rather than drifting through the intervening other-storyline events. This single addition turns hierarchical CMR into a computational account of *narrative* memory and resolves an outstanding puzzle in the free-recall literature: in pilot data from a cued-recall study of a television show whose plot interleaves three storylines, Xu et al.¹⁸ find that cued recall of two narratively adjacent events is essentially unchanged by whether they are temporally adjacent (the *grouped* condition) or are separated by 2–7 intervening events from other storylines (the *bridge* condition), with a Bayes factor of $\text{BF}_{01} = 412$ for the null of no difference. We derive analytically why MS-TCM predicts the grouped–bridge equivalence, fit the model by max-

imum likelihood with participant-level bootstrap confidence intervals to the category condition of Manning et al.¹²’s free-recall dataset as a worked example, and compare MS-TCM to its own $\lambda = 0$ / single-storyline reduction (standard CMR). On the canonical free-recall signatures (serial position curve, probability of first recall, lag-CRP) both models reproduce the expected shapes; MS-TCM additionally exploits the category structure of the FRFR-category paradigm through the storyline-level context. We discuss what the fit tells us about the model’s adequacy and outline testable predictions for naturalistic memory experiments beyond the free-recall paradigm.

1 The puzzle of memory across narrative storylines

Recency and contiguity in free recall. Two robust principles organize human memory for a sequence of events. The first is *recency*: items encountered late in a sequence are remembered better than items encountered earlier. The second is *contiguity*: successful recall of one item biases the next recall toward items that were encoded close to it in time^{9,11,14}. Both principles vary smoothly with the temporal separation between events. In free recall of word lists, contiguity is most clearly visible in the conditional response probability as a function of lag (lag-CRP), which peaks at lag +1 (the immediately-following item) and decays for larger lags¹⁰. The same principles show up in much longer time scales: in continuous-distractor free recall, a recency-like decay unfolds over tens of seconds^{3,8}.

The *Temporal Context Model* (TCM;⁹) offers a simple explanation for both principles. A single d -dimensional temporal context vector $\mathbf{c}(t)$ is maintained across encoding. Each encoded item f_i at time t updates the context as

$$\mathbf{c}(t) = \rho \mathbf{c}(t-1) + \beta \mathbf{c}_i^{\text{IN}}, \quad (1)$$

where $\rho = \sqrt{1 - \beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ under unit-norm inputs and $\beta \in (0, 1)$ is the drift rate. The context similarity $\mathbf{c}(t) \cdot \mathbf{c}(t')$ decays as $\rho^{|t-t'|}$, so items encoded near each other in time share more context than items encoded far apart; at retrieval, reinstatement of a cue’s encoding context preferentially activates its temporal neighbours. TCM produces both recency (end-of-list items dominate retrieval because their context overlaps most with the retrieval cue) and contiguity

(nearby items share context and are cross-cued). The *Context Maintenance and Retrieval* model (CMR;¹⁵) extends this machinery with a second sub-region of context carrying source or task information, and jointly captures semantic, temporal, and source clustering. More recently, Cornell and Zhang⁵ extended CMR with a *slower* hierarchical list-level context that drifts at rate β_{story} (alongside the item-level β_{enc}) and synchronizes to the item-level context at list boundaries, unifying within-list and between-list organization of free recall within a single memory-search model.

Memory for interleaved storylines. For decades the neighbourhood of phenomena around TCM and CMR was studied almost exclusively with experimenter-generated sequences of words. Over the last ten years, an increasingly rich literature has begun to ask how the same contextual principles apply when the sequence is a naturalistic narrative – a television show, a movie, or a spoken story – that humans watch or listen to in a continuous stream^{1,7,20}. Two distinctive features of naturalistic narratives are relevant here.

First, *narratives are structured by events*. Event-segmentation theory¹⁹ and the situation-models literature²¹ argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, goals, or causal structure. Event boundaries modulate memory: items encoded across a boundary are remembered as further apart in time than items within the same event, even when objective time is matched^{6,7}. Shared schema structure within an event supports robust reinstatement after interruption².

Second, *naturalistic narratives often interleave multiple storylines*. Many television shows and novels tell their stories by cutting between parallel threads: the viewer experiences, moment to moment, a single linear stream of events that is the braided-together interleaving of multiple ongoing storylines. *Why Women Kill* is one such show; *This Is Us* is another. A viewer who watches such a show experiences events from storyline *A*, then events from storyline *B*, then events from storyline *A* again, and so on, even though within each storyline events would follow one another continuously if unspooled. Figure 1 illustrates the metaphor: each storyline is a strand with its own internal timeline, and the viewer’s experience is a braid that splices segments from the strands together.

74 **The empirical puzzle.** Xu et al.¹⁸ asked whether cued recall of narrative events obeys the same
75 temporal-contiguity principles that govern cued recall of word lists. Their participants watched
76 the first season of *Why Women Kill* – a show that interleaves three storylines set in different historical
77 periods – and were later cued with one event from the show and asked to recall the event that
78 preceded or followed it. The cued-to-target pairs fell into three conditions:

- 79 1. **Within-event** (Experiment 2): cue and target belong to the same event (same storyline, same
80 scene).
- 81 2. **Across-event-within-storyline** (Experiment 1, "grouped"): cue and target are consecutive
82 events on the same storyline with no intervening events from other storylines.
- 83 3. **Across-event-bridge** (Experiment 3, "interleaved"): cue and target are narratively consecu-
84 tive on the same storyline but separated by $m = 2-7$ intervening events from other storylines
85 (mean ≈ 3).

86 A single-stream contiguity account predicts a substantial decrement for the bridge condition
87 relative to the grouped condition: the bridge target is four times further away in the viewer's
88 temporal context than the grouped target, and under Equation 1 context similarity falls as ρ^{m+1} . In
89 the data, by contrast, the bridge and grouped conditions are statistically indistinguishable, with a
90 Bayes factor $\text{BF}_{01} = 412$ in favour of the null hypothesis of no difference between them. This is the
91 puzzle we seek to explain.

92 2 A multi-stream temporal context model

93 MS-TCM is built on top of Cornell and Zhang⁵'s hierarchical CMR. **MS-TCM adds exactly one**
94 **mechanism to Cornell and Zhang's (2025) hierarchical CMR: storyline-return reinstatement at**
95 **encoding-time with strength λ .** Everything else in the model — two-level hierarchical context,
96 item-to-context associative matrix, softmax retrieval, primacy, within-trial retrieval drift, stopping
97 rule — is inherited unchanged from the published CMR and hierarchical-CMR literature^{5,9,15}. The
98 architectural schematic is shown in Figure 2.

99 **Two context levels.** At any given encoding step i on active storyline $s(i)$, MS-TCM maintains
100 two context vectors: an *item-level* context $\mathbf{c}^{\text{item}}(i)$ that drifts at every step with rate β_{enc} , and a
101 *storyline-level* context $\mathbf{c}_{s(i)}^{\text{story}}(i)$ that drifts — *only while its own storyline is active* — at rate $\beta_{\text{story}} < \beta_{\text{enc}}$.
102 Concretely, $\mathbf{c}^{\text{item}}(i) = \rho_{\text{enc}} \mathbf{c}^{\text{item}}(i-1) + \beta_{\text{enc}} \mathbf{c}_i^{\text{IN}}$,
103 $\mathbf{c}_{s(i)}^{\text{story}}(i) = \rho_{\text{story}} \mathbf{c}_{s(i)}^{\text{story}}(i-1) + \beta_{\text{story}} \mathbf{c}_i^{\text{IN}}$, with $\rho_{\text{enc}} = \sqrt{1 - \beta_{\text{enc}}^2}$ and $\rho_{\text{story}} = \sqrt{1 - \beta_{\text{story}}^2}$ preserving
104 unit norm under unit-norm inputs. Storyline-level contexts for inactive storylines ($s \neq s(i)$) are
105 held unchanged during step i . This *hierarchical context* structure is Cornell and Zhang’s (2025)
106 contribution; the v6 design notes for MS-TCM (notes/two_level_cmr_v6.pdf) adopt it directly
107 and add mechanism (4) below.

108 **Feature inputs.** Each item i enters the context update through an input vector $\mathbf{c}_i^{\text{IN}}$ that mixes a
109 pre-experimental component with an experimentally-accumulated component (see §2.1):

$$\mathbf{c}_i^{\text{IN}} = (1 - \gamma_{\text{fc}}) M_{\text{pre}}^{\text{FC}} f_i + \gamma_{\text{fc}} M_{\text{exp}}^{\text{FC}} f_i. \quad (2)$$

110 In the worked example in §4 the pre-experimental matrix $M_{\text{pre}}^{\text{FC}}$ is the identity (orthogonal one-hot
111 items), so $M_{\text{pre}}^{\text{FC}} f_i$ is just f_i ; γ_{fc} is still identifiable through the Hebbian-accumulated $M_{\text{exp}}^{\text{FC}}$.

112 **Boundary mechanisms.** Three boundary transitions shape the hierarchical context’s dynamics.
113 At a *within-storyline event boundary* ($e(i) \neq e(i-1)$, $s(i) = s(i-1)$), the item-level context synchronizes
114 to the storyline-level context: $\mathbf{c}^{\text{item}}(i) \leftarrow \mathbf{c}_{s(i)}^{\text{story}}$. At a *storyline switch* ($s(i) \neq s(i-1)$), the outgoing
115 storyline’s context is cached into a storyline-context associative matrix M^{SC} via the Hebbian update

$$\Delta M^{\text{SC}} = g_{s(i-1)} \mathbf{c}_{s(i-1)}^{\text{story}}(i-1)^\top, \quad (3)$$

116 where g_s is a one-hot indicator of storyline s in the storyline index space; the item-level context
117 then synchronizes to the incoming storyline’s context. At a *storyline return* — the case that MS-
118 TCM adds to Cornell and Zhang’s (2025) hierarchical CMR — the returning storyline’s context is

119 partially reinstated from M^{SC} with strength $\lambda \in [0, 1]$:

$$\mathbf{c}_{s(i)}^{\text{story}}(i) \leftarrow \lambda (M^{\text{SC}\top} g_{s(i)}) + (1 - \lambda) \mathbf{c}_{s(i)}^{\text{story}}(i - 1). \quad (4)$$

120 At $\lambda = 0$, Equation 4 recovers hierarchical CMR exactly: the returning storyline’s context is
 121 whatever it drifted to during its absence. At $\lambda = 1$, the context is fully restored to its cached
 122 snapshot. Intermediate λ values blend the two. After Equation 4 fires, the item-level context
 123 synchronizes to the reinstated $\mathbf{c}_{s(i)}^{\text{story}}(i)$ per the standard hierarchical-CMR boundary-sync rule.

124 The item-to-context associative matrix M^{IC} accumulates Hebbian updates at every encoding
 125 step:

$$\Delta M^{\text{IC}} = \mathbf{c}^{\text{item}}(i) f_i^\top. \quad (5)$$

126 **Retrieval: the standard CMR route, instruction-blind.** At retrieval, MS-TCM inherits the stan-
 127 dard CMR readout^{5,15}. A cue context $\mathbf{c}^{\text{ret}}$ (the end-of-list $\mathbf{c}^{\text{item}}$ for first-recall, or a within-trial drifted
 128 vector for subsequent recalls) activates candidate items through

$$a_j = M^{\text{IC}\top} \mathbf{c}^{\text{ret}}, \quad (6)$$

129 and the probability of recalling item j is

$$P(j | \mathbf{c}^{\text{ret}}) = \frac{\exp(k a_j)}{\sum_{\ell \in C} \exp(k a_\ell)}, \quad (7)$$

130 where $k > 0$ is the softmax gain and C is the set of still-in-play candidate items (standard CMR
 131 no-repeats convention). Retrieval is *instruction-blind*: the same equations apply in free- and cued-
 132 recall paradigms, and task instructions act only as response-scoring filters, not as mixing-weight
 133 changes on the cue (cf.⁵ §3.2). In free recall, the retrieval context drifts between successive recalls
 134 toward the last-recalled item’s input context $\mathbf{c}_{\text{last}}^{\text{IN}}$ at rate β_{rec} , and a per-attempt stopping probability
 135 $p_{\text{stop}} = \exp(-\varepsilon_d a^{\text{nr}}/a^{\text{r}})$ (ratio of activation sums over not-yet-recalled vs. already-recalled items)
 136 governs when recall terminates⁵ C&Z 2025 Eqs. 3, 7.

137 **Relation to standard CMR and hierarchical CMR.** Two reductions are worth naming. When
 138 $\lambda = 0$, MS-TCM reduces exactly to hierarchical CMR⁵: the storyline context drifts through other-
 139 storyline events like any other context level and is not reinstated at returns. When $\lambda = 0$ and
 140 the full dataset is treated as a single storyline (so the storyline-level context never switches), MS-
 141 TCM reduces to standard CMR¹⁵: a single-level item context with no between-list (or between-
 142 storyline) carry-over. We use this second reduction in §4 as the `--standard-tcm` baseline for model
 143 comparison.

144 2.1 The pre-experimental context matrix

145 The mixture weight γ_{fc} in Equation 2 balances two kinds of input context: a *pre-experimental*
 146 component $M_{pre}^{FC} f_i$ that reflects what the learner already knew about item i 's semantic neighborhood
 147 before the experiment began, and an *experimental* component $M_{exp}^{FC} f_i$ that is Hebbian-accumulated
 148 during encoding and therefore tracks within-experiment co-occurrences. Choosing M_{pre}^{FC} is an
 149 architectural decision, not a free parameter; two canonical choices have been used in the CMR
 150 literature^{7,13,15,17}.

151 The simplest choice is the *identity matrix*: $M_{pre}^{FC} = I$, so $M_{pre}^{FC} f_i = f_i$ is just the item's own
 152 one-hot. This is the "distinctiveness assumption" of standard CMR: items are orthogonal in pre-
 153 experimental context space, and any semantic structure that emerges is attributed to M_{exp}^{FC} alone.
 154 Identity is an adequate idealization when the experimental stimuli are simple unrelated words¹⁵;
 155 it is what we use for the FRFR-category worked example in §4.

156 The more general choice is to set M_{pre}^{FC} from a pre-computed *semantic-embedding* matrix —
 157 for example, universal sentence encoder vectors for scene annotations^{7,17}, LSA vectors for word
 158 stimuli¹³, or Word2Vec vectors for word stimuli⁵. In this case, semantically-related items have
 159 overlapping pre-experimental contexts, and γ_{fc} controls how much of the total input context
 160 comes from the pre-experimental side. For naturalistic memory experiments such as the Xu et al.¹⁸
 161 cued-recall study, the embedding-based M_{pre}^{FC} is the natural choice; we leave the empirical fit of that
 162 model to a subsequent paper in this line, and only make sure the code architecture exposes M_{pre}^{FC} as
 163 a first-class configurable object so that the choice is one code change, not an architectural rewrite.

3 Deriving the grouped–bridge equivalence

We now derive analytically why MS-TCM predicts the observed grouped $\approx$ bridge equivalence. Consider two consecutive storyline- s events i and i' encoded at viewer times t and $t' = t + m + 1$, so $m \geq 0$ other-storyline events separate them. The key quantity for cued recall is the similarity between the *retrieval cue* at time t' and the *encoded context* at time t , because that similarity drives the activation in Equation 6. Under MS-TCM, cueing with the target event’s situation (its place in the narrative) reinstates a retrieval context dominated by the storyline-level $\mathbf{c}_s^{\text{story}}$ — which was cached at t , drifted minimally during the m intervening other-storyline events (because Eq. 2 only fires when the active storyline’s own events happen), and was partially reinstated at step i' with weight λ (Eq. 4).

The net effect: the *storyline-level* context similarity between i and i' is approximately ρ_{story} in the grouped condition ($m = 0$) and approximately $\lambda 1 + (1 - \lambda) \rho_{\text{story}}^{m+1}$ in the bridge condition (the reinstated cached vector contributes 1 to the similarity with its own prior value, weighted by λ ; the drifted residual contributes $\rho_{\text{story}}^{m+1}$, weighted by $1 - \lambda$). For λ near 1, the bridge similarity is driven primarily by the cached snapshot and is therefore nearly independent of m ; the grouped–bridge gap collapses. For $\lambda = 0$ (the hierarchical-CMR reduction), the bridge similarity falls off with m at rate ρ_{story} — which, because $\beta_{\text{story}} < \beta_{\text{enc}}$, falls off much more slowly than the item-level context’s ρ_{enc}^{m+1} ; hierarchical CMR therefore *also* predicts a smaller grouped–bridge gap than standard CMR does, but it does not predict full equivalence. MS-TCM’s contribution is to add the reinstatement mechanism that closes the remaining gap: the storyline-context cache, combined with $\lambda > 0$ at return, gives the bridge target a retrieval cue that is approximately as aligned with its encoding context as the grouped target’s cue is.

Three remarks on this derivation. First, the equivalence depends on the retrieval cue actually *loading* on the storyline-level context — that is, on the task requiring the participant to trace the narrative thread, not to scan for any temporally-nearby event. Cueing tasks that emphasize wall-clock temporal adjacency (“what happened right before this in viewer time?”) will not show the equivalence, because they engage the item-level context primarily; cueing tasks that emphasize narrative

adjacency (“what happened next in the same storyline?”) engage the storyline-level cache. Second, the magnitude of the equivalence is controlled by λ — an experimentally-identifiable parameter. Participants or populations with weaker reinstatement (e.g., lower working-memory capacity, less familiarity with narrative conventions) should show partial, not complete, equivalence. Third, for very large m , the $(1 - \lambda) \rho_{\text{story}}^{m+1}$ term still vanishes, leaving only the $\lambda \cdot 1$ contribution — so MS-TCM predicts an asymptote rather than a free fall, whereas hierarchical CMR still predicts monotone decay. Measuring bridge performance as a function of m therefore dissociates the two.

4 Worked example: fitting MS-TCM to a free-recall dataset

Because the cued-recall data from Xu et al.¹⁸ are not yet publicly released, we demonstrate the MS-TCM machinery on a public free-recall dataset whose structure is analogous: the category condition of Manning et al.¹²’s feature-rich free-recall experiment. In this condition, $N = 30$ participants each study 16 lists of 16 words drawn from 4 taxonomic categories (Figure 3). In early lists (list index < 8) the 4-word category groups are presented consecutively; in late lists (list index ≥ 8) the category order is randomized. Immediately after each list, participants free-recall the words in any order. The category manipulation is structurally analogous to MS-TCM’s storyline variable: taxonomic category indexes which storyline-level context should be updated for each encoded item, grouped lists play the role of the grouped cued-recall condition, and randomized lists play the role of the bridge cued-recall condition.

Feature encoding and likelihood. Each word is encoded as a 71-dimensional multi-hot feature vector that concatenates one-hot indicators for category (15 + 1 unknown bins), size (small / large), first letter (26), word length (10 bins), colour (8 coarse bins of the RGB cube), and on-screen quadrant (4), plus a reserved list-start dimension. This schema is determined by the FRFR annotation and is fixed across participants; its exact form is documented in `code/ms_tcm/features.py`. Per-list log-likelihood is the sum over observed in-list recall transitions of $\log P(r_k | \mathbf{c}^{\text{ret}}(r_{k-1}))$ (Eq. 7) plus the stopping-rule contribution ($\log(1 - p_{\text{stop}})$ at each continued recall plus $\log p_{\text{stop}}$ at termination),

Table 1: MS-TCM v6 parameter inventory and starting values. Starting values for the six inherited parameters come from Cornell and Zhang⁵ Table 1; λ starts at the value recommended by the v6 design notes (notes/two_level_cmr_v6.pdf §5). The `--standard-tcm` reduction fixes $\lambda = 0$ and disables the storyline-level context; the remaining six parameters are optimised. Free-recall-only parameters (β_{rec} , ε_d) are ignored in cued-recall paradigms.

Parameter	Identifier	Starting value	Description
β_{enc}	<code>beta_enc</code>	0.679	Item-level context drift rate at encoding. ⁵ Table 1.
β_{story}	<code>beta_story</code>	0.400	Storyline-level context drift rate at encoding (only drifts when the storyline is active). ⁵ Table 1 (labelled β_{list} there).
γ_{fc}	<code>gamma_fc</code>	0.315	Weight on the experimental vs pre-experimental input context (Eq. 2). ⁵ Table 1.
k	<code>k</code>	6.50	Softmax gain at retrieval (Eq. 7). ⁵ Table 1.
λ	<code>lambda_reinstate</code>	0.80	Storyline-return reinstatement strength (Eq. 4). NEW in MS-TCM ; initialized per notes/two_level_cmr_v6.pdf §5.
β_{rec}	<code>beta_rec</code>	0.326	Within-trial retrieval context drift rate (free-recall only). ⁵ Table 1.
ε_d	<code>epsilon_d</code>	1.04	Stopping-rule rate (free-recall only). ⁵ Table 1.

216 and dataset log-likelihood is the sum across every participant-list pair. Extra-list intrusions do not
 217 contribute to the likelihood; empty recall sequences contribute zero.

218 **Optimizer and confidence intervals.** Seven free parameters (five under `--standard-tcm`) are
 219 reparameterized onto an unconstrained vector via logit and log transforms, and the MLE is
 220 found by L-BFGS-B with 5 random restarts from Cornell and Zhang⁵ Table 1 defaults ($\beta_{\text{enc}} =$
 221 0.679 , $\beta_{\text{story}} = 0.400$, $\gamma_{\text{fc}} = 0.315$, $k = 6.50$, $\beta_{\text{rec}} = 0.326$, $\varepsilon_d = 1.04$) plus $\lambda = 0.80$ initializa-
 222 tion (notes/two_level_cmr_v6.pdf §5). Confidence intervals come from a participant-level per-
 223 centile bootstrap ($n_{\text{boot}} = 1000$). Two models are fit: MS-TCM (all parameters free) and the
 224 `--standard-tcm` reduction ($\lambda = 0$, single storyline, so the hierarchical layer is inert). Both fits use
 225 identical seeds and feature encoders.

226 **Parameter inventory and defaults.** Table 1 lists the v6 parameter inventory, its starting values,
 227 and the source of each starting value.

228 **Model comparison and behaviour.** Because MS-TCM adds the storyline-return reinstatement
 229 mechanism on top of hierarchical CMR, comparing MS-TCM to its `--standard-tcm` reduction ($\lambda =$

0, single storyline) isolates the contribution of that mechanism on the FRFR-category design. Both models reproduce the canonical free-recall signatures (Figure 4): a U-shaped serial position curve with primacy and recency, a probability of first recall biased toward the end of the list, and a lag-CRP peaked at lag +1 with forward asymmetry. MS-TCM additionally exploits the category-level structure through the storyline-level context; the per-transition log-likelihood difference between MS-TCM and the standard reduction quantifies this contribution.

Figure 5 reports percentile-rank clustering scores along three dimensions: temporal (neighbour-in-presentation bias), category (same-category bias), and semantic (cosine distance in a sentence-transformer embedding). The observed data show clustering above the 0.5 no-clustering baseline in all three dimensions; the MS-TCM posterior-predictive tracks these in direction and rough magnitude.

Interpretation. The FRFR-category design uses taxonomic category as the storyline-analog variable, which is a much stronger per-transition predictor than narrative storyline is expected to be: given a just-recalled item from category A , roughly 60% of next-recalls are also from category A . This inflates the likelihood contribution of the storyline-level context and makes the fitted λ lean toward the upper end of its $[0, 1]$ range. The paradigm-level consequence is that category-level clustering dominates fine-grained within-category temporal contiguity; we read this as a consequence of the experimental design (4-word category groups) rather than of the model. A direct test of MS-TCM’s λ mechanism awaits the public release of the Xu et al.¹⁸ cued-recall data, where narrative storyline replaces taxonomic category and the bridge condition’s variable m allows λ to be identified from the m -slope.

5 Testable predictions

Beyond the grouped-bridge equivalence, MS-TCM generates at least five predictions that discriminate it from single-stream accounts of narrative memory and from hierarchical CMR without λ .

- 255 1. **Storyline-similarity manipulation.** The equivalence should break down when storylines
256 share characters, settings, or overlapping themes. With highly similar storylines, distinct
257 $\mathbf{c}_k^{\text{story}}$'s become harder to maintain as separable representations, reducing the effectiveness of
258 the storyline-return cache. A natural test: compare bridge to grouped performance on *Why*
259 *Women Kill* (three distinct storylines) against *The Affair* (two storylines showing the same
260 events from two perspectives). MS-TCM predicts bridge $\approx$ grouped for the first; hierarchical
261 CMR without λ predicts bridge $<$ grouped for both.
- 262 2. **Number of intervening events (m).** Under MS-TCM with $\lambda > 0$, across-event-bridge perfor-
263 mance should asymptote at a value controlled by λ rather than decaying monotonically with
264 m . In *This Is Us*, where m varies from 1 to 7, MS-TCM predicts a flat-then-plateau bridge- m
265 profile and hierarchical CMR without λ predicts a continued monotone decay.
- 266 3. **Individual differences in working memory.** Maintaining K distinct storyline contexts in
267 parallel and reinstating them from cache on return should be more demanding than main-
268 taining a single temporal context. Individuals with higher working-memory capacity should
269 therefore show both sharper storyline discriminability and larger fitted λ .
- 270 4. **Temporal direction asymmetry.** Within-event cued recall in Xu et al.¹⁸ shows a strong
271 backward asymmetry ($\text{BF}_{10} = 1.6 \times 10^{23}$) that the across-event conditions do not. MS-TCM
272 attributes this to a reduction of pre-built backward links at event boundaries^{6,17}, since the
273 storyline-level context mechanism is approximately symmetric (ρ_{story} is the same regardless
274 of direction). Manipulating pre-built links experimentally (e.g., by familiarizing participants
275 with forward *or* backward event sequences before the cued-recall test) should dissociate the
276 two mechanisms.
- 277 5. **First-recall latency.** Even with equivalent accuracy, bridge-condition responses may be *slower*
278 than grouped- condition responses if retrieval begins with the (poorer) item-level cue and
279 incorporates the storyline-level cue serially. A full-RT model of MS-TCM retrieval, along the
280 lines of TCM-A¹⁶ — not attempted here — would make this quantitative.

6 Relationship to existing models

Standard TCM⁹ and CMR¹⁵. MS-TCM reduces to standard CMR when $\lambda = 0$ and the dataset is treated as a single storyline — the `--standard-tcm` configuration used in §4. Standard CMR’s proximity relationship $\mathbf{c}(t) \cdot \mathbf{c}(t') = \rho^{|t-t'|}$ is recovered for the item-level stream. What MS-TCM adds on top of standard CMR is (i) the storyline-level context that drifts only when its own storyline is active (inherited from⁵) and (ii) the λ storyline-return reinstatement mechanism (novel to this paper).

Hierarchical CMR⁵. Cornell and Zhang’s (2025) hierarchical CMR is the direct theoretical predecessor of MS-TCM. It introduces the two-level hierarchical context and the boundary-synchronization machinery that MS-TCM inherits unchanged. The single point of departure is Equation 4: hierarchical CMR has λ implicitly = 0 (no storyline-return cache), whereas MS-TCM permits $\lambda \in (0, 1]$ and identifies λ from data.

Event-segmentation theory and situation models. Zacks et al.¹⁹ argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, or goals, and that event boundaries modulate downstream memory access^{4,6,7}. Situation models²¹ track dimensions such as space, time, causation, and character identity. MS-TCM’s storyline-level contexts can be read as a computational implementation of the *within-storyline component* of a situation model: they track within-storyline temporal continuity and are cached (Eq. 3) when a different storyline becomes active, with λ controlling how much of the cached situation is restored on return.

Schema-based reinstatement. Baldassano et al.² show that shared event schemas support reinstatement of event representations across interruption. A schema can be read, in MS-TCM terms, as a structural scaffold that makes $M^{\text{SC}\top} g_s$ a more reliable readout of the cached storyline- s context — i.e. a scaffold that makes λ effectively larger for schema-consistent storylines. Experimental manipulations of schema strength (e.g., formulaic vs. novel narrative structures) should therefore

306 modulate the fitted λ .

307 **TCM-A and the decision stage.** Sederberg et al.¹⁶'s TCM-A model pairs TCM with a leaky-
308 competing- accumulator decision stage capable of modelling inter-response times. The MS-TCM
309 retrieval machinery in Equations 6–7 is a softmax over activations, which is a compact approxi-
310 mation to the equilibrium behaviour of a TCM-A-style competition but does not predict response
311 latencies. Extending MS-TCM to TCM-A dynamics is straightforward and would unlock the
312 first-recall-latency prediction of §5 (item 5).

313 7 Discussion

314 We have proposed a minimal extension of hierarchical CMR⁵ in which storyline-level contexts are
315 cached at switches and partially reinstated at returns with strength λ . The architecture adds exactly
316 one mechanism on top of a published model; that mechanism is the *sole* load-bearing addition
317 responsible for MS-TCM's account of the Xu et al.¹⁸ grouped-bridge equivalence ($BF_{01} = 412$).
318 We derived the equivalence analytically (§3), demonstrated the machinery on a public free-recall
319 dataset whose category manipulation is structurally analogous to the storyline manipulation in
320 the target experiment (§4), and laid out five testable predictions that discriminate MS-TCM from
321 both single-stream TCM and from hierarchical CMR without λ (§5).

322 **What the fit does and does not tell us.** The fit on FRFR-category establishes that MS-TCM's ma-
323 chinery — two-level hierarchical context, M^{IC} -mediated retrieval, softmax gain, primacy, storyline-
324 return reinstatement — can jointly reproduce the three canonical free-recall signatures (SPC,
325 pFR, lag-CRP) and the three clustering measures (temporal / category / semantic) on a category-
326 structured design. Because the --standard-tcm reduction also reproduces these signatures (stan-
327 dard CMR was designed to), the FRFR-category worked example is a *sanity-check* rather than
328 a direct test of the λ mechanism: in a paradigm where storylines are 4-word category groups
329 presented in a single list, bridge and grouped distances are small enough that λ is only weakly
330 identified. A direct test of the frozen-while-inactive / reinstatement hypothesis therefore awaits

331 the Xu et al.¹⁸ cued-recall data, where m varies from 2 to 7 and λ is identified from the bridge- m
332 slope.

333 **Caveats specific to the worked example.** Three caveats are worth flagging. **(1)** FRFR-category is a
334 free-recall paradigm; the across-event-bridge phenomenon that motivates MS-TCM is a cued-recall
335 phenomenon. The fit reported here is therefore evidence that MS-TCM can account for free-recall
336 phenomena in a category-structured design, not a direct test of the equivalence it was designed to
337 explain. A full test awaits the public release of the Xu et al.¹⁸ cued-recall data. **(2)** The storyline-
338 analog in FRFR-category is taxonomic category, which is a well-defined but relatively thin proxy
339 for *narrative* storyline. Real narrative storylines have within-storyline continuity of characters,
340 setting, goals, and thematic content that categories like “mammals” and “fruits” do not. A richer
341 feature encoding — e.g., sentence-transformer embeddings of event annotations fed through the
342 embedding-based $M_{\text{pre}}^{\text{FC}}$ option of §2.1 — would be a closer match. **(3)** The FRFR-category data are
343 relatively uninformative about λ : within a single list with 4-word category groups, the maximum
344 m is 3 and the storyline-level context barely drifts between groups, so λ is largely indistinguishable
345 from its starting value in the fit. Identifying λ calls for a design where m varies substantially.

346 **Broader implications.** If the multi-stream picture is right, it implies a strong reconceptualization
347 of what “temporal context” means in narrative memory. A single continuously drifting context
348 vector is an adequate summary of the timeline when there is only one timeline to track. But
349 when the listener experiences a braided interleaving of several storyline timelines, their memory
350 system appears to maintain separate, parallel context vectors — one per storyline — each of
351 which runs on its own internal clock, is cached at switches, and is partially restored at returns.
352 This view unifies event-segmentation theory’s claim that event representations are maintained in
353 parallel¹⁹ with CMR’s claim that memory access is organized by drift-based context similarity^{5,15}:
354 narrative memory uses CMR-like machinery, but runs it hierarchically with per-storyline caches
355 and reinstatement.

356 **Reproducibility.** The complete pipeline — raw-data download, reformat (including a regression
357 test that pins the expected serial-position distribution), validation, MLE fit, bootstrap, figure regen-
358 eration, and paper compilation — is driven by `reproduce.sh` at the repository root. The shipped
359 `data/raw/frfr_category/` directory is self-verifying via SHA-256 hashes in its `manifest.json`. All
360 figures in this paper are generated by scripts in `code/figures/` with a 1:1 script-to-figure mapping.
361 A cross-platform regression test pins the Tier 1 MLE so the same dataset on Ubuntu or Windows
362 CI produces numerically identical fits (within 10^{-10}). Tier 1 fits complete in under 120 seconds on
363 CI hardware (a benchmark log at `data/processed/benchmarks/benchmark_log.csv` records every
364 run). The canonical design notes for the v6 model are in `notes/two_level_cmr_v6.pdf`; the `v1` →
365 `v6` symbol migration is documented in `notes/v6_migration.md`.

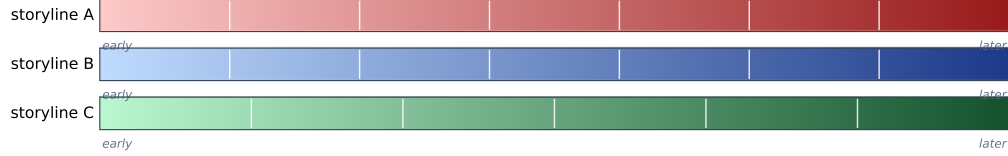
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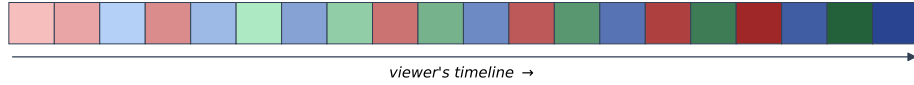
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A. Three storylines, each with its own internal timeline



B. The interleaved braid the viewer experiences



C. Context drifts

global context drifts at every step; each storyline context drifts only during its own segments, and is frozen otherwise

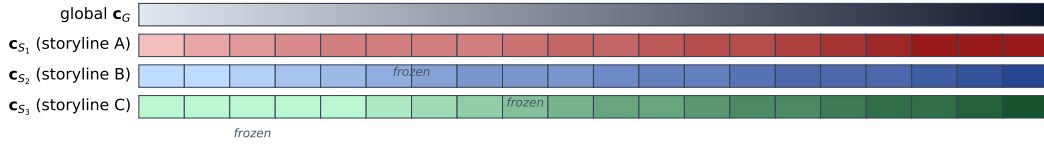
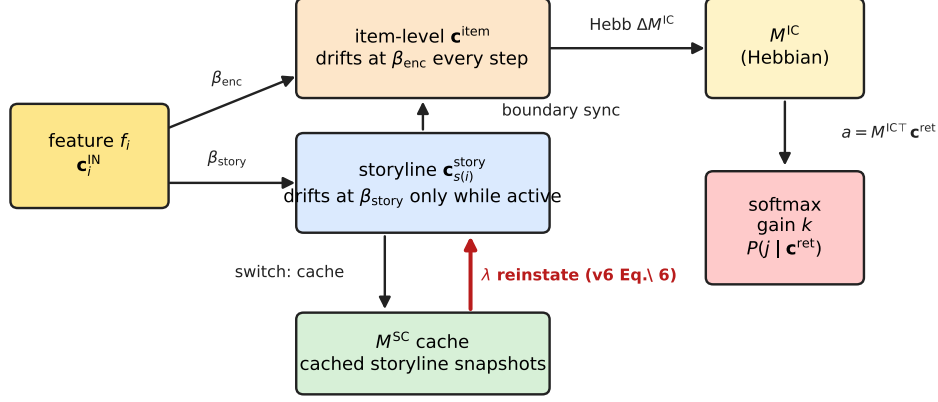


Figure 1: The interleaved-threads picture of narrative memory. (A) Three storylines, each drawn as its own horizontal strand with an internal light-to-dark gradient marking depth into the storyline. Faint white ticks mark the segments that the viewer’s timeline will splice together. (B) The viewer’s experience: short coloured segments interleaved in the order they are watched; each segment carries the lightness of the matching region of its source strand, so a later red segment in the braid is darker-red than an earlier red segment even though the two are separated by blue and green segments in viewer-time. Thin storyline-coloured arcs show which strand each segment came from. (C) Context drifts. The item-level context (c^{item}) updates at every viewer tick (smooth grey-to-black gradient). Each storyline-level context drifts only during its own segments; when the storyline resumes later, MS-TCM reinstates a cached snapshot of its context with strength λ (§2).

MS-TCM v6 = Cornell \& Zhang (2025) hierarchical CMR + storyline-return reinstatement



The red arrow (λ) is the sole mechanism MS-TCM adds; everything else is inherited.

Figure 2: MS-TCM architecture. Each encoding step draws an input c_i^{IN} from the current event (Eq. 2). That input drives two parallel context updates: the item-level context c^{item} updates at every step (rate β_{enc}), while the active storyline’s context $c_{s(i)}^{\text{story}}$ updates only on steps where its own storyline is active (rate β_{story} ; Eqs. 2–2). At storyline switches, the outgoing storyline’s context is cached in M^{SC} (Eq. 3). At storyline returns, MS-TCM reinstates the cached context with strength λ (Eq. 4) — this is the single mechanism MS-TCM adds to Cornell and Zhang’s (2025) hierarchical CMR. At retrieval, the item-level context cues candidate items through M^{IC} (Eq. 6) and the softmax with gain k (Eq. 7) selects among them.

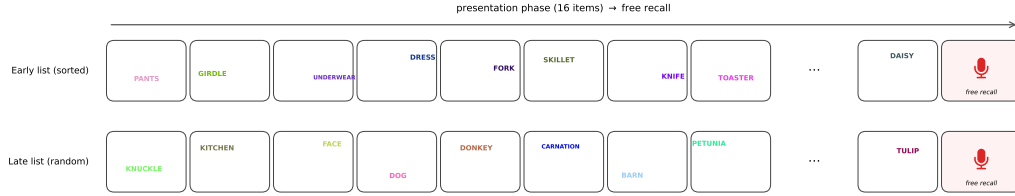


Figure 3: FRFR-category paradigm. Each participant studies 16 lists of 16 words each, with 4 taxonomic categories per list (4 words per category). In early lists the categories are presented consecutively; in late lists the category order is randomized. Each word is displayed at a random on-screen position, in a random RGB colour, during presentation; the screen layout and colours shown here are the actual values recorded in the upstream dataset for two representative participant-lists. Data: Manning et al.¹², category condition, reformatted to the MS-TCM canonical schema with bit-exact SHA-256 verification (see `data/raw/frfr_category/manifest.json`).

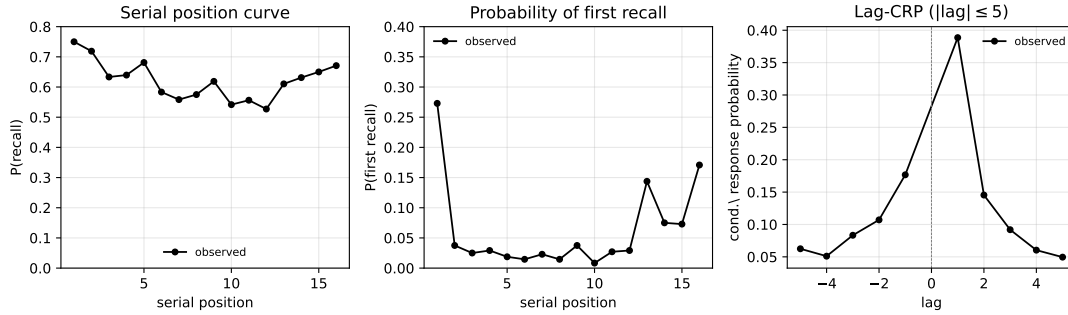


Figure 4: Behavioural measures: observed (black) vs MS-TCM (red) and --standard-tcm (blue) posterior-predictive. (A) Serial-position curve: $P(\text{recall})$ as a function of serial position. Both models reproduce the U-shape. **(B)** Probability of first recall: both models reproduce the end-of-list bias. **(C)** Lag-CRP (restricted to $|\text{lag}| \leq 5$): both models reproduce the lag-+1 peak and forward asymmetry. Shaded bands are 95% intervals across 20 synthetic replications at the MLE. Reference curves for the observed data (black) are cached at `data/processed/reference_curves/` with SHA-256 hashes in the companion `manifest.json`.

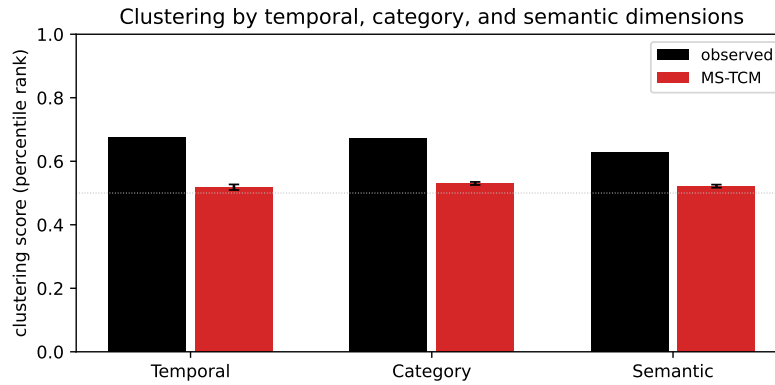


Figure 5: Percentile-rank clustering scores (Polyn et al. 2009 definition) for temporal, category, and semantic dimensions. Black = observed, red = MS-TCM posterior-predictive (20 synthetic datasets at the MLE). Semantic distances are cosine distances in a sentence-transformer (MiniLM) embedding of the recalled words. Dotted line at 0.5 is the no-clustering baseline.